

Think OS

A Brief Introduction to Operating Systems

Version 0.7.4

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Allen B. Downey

Green Tea Press

Needham, Massachusetts

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Green Tea Press
9 Washburn Ave
Needham MA 02492

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The L^AT_EX source for this book is available from <http://greenteapress.com/thinkos>.

Preface

In many computer science programs, Operating Systems is an advanced topic. By the time students take it, they know how to program in C, and they have probably taken a class in Computer Architecture. Usually the goal of the class is to expose students to the design and implementation of operating systems, with the implied assumption that some of them will do research in this area, or write part of an OS.

This book is intended for a different audience, and it has different goals. I developed it for a class at Olin College called Software Systems.

Most students taking this class learned to program in Python, so one of the goals is to help them learn C. For that part of the class, I use Griffiths and Griffiths, *Head First C*, from O'Reilly Media. This book is meant to complement that one.

Few of my students will ever write an operating system, but many of them will write low-level applications in C or work on embedded systems. My class includes material from operating systems, networks, databases, and embedded systems, but it emphasizes the topics programmers need to know.

This book does not assume that you have studied Computer Architecture. As we go along, I will explain what we need.

If this book is successful, it should give you a better understanding of what is happening when programs run, and what you can do to make them run better and faster.

Chapter 1 explains some of the differences between compiled and interpreted languages, with some insight into how compilers work. Recommended reading: *Head First C* Chapter 1.

Chapter 2 explains how the operating system uses processes to protect running programs from interfering with each other.

Chapter 3 explains virtual memory and address translation. Recommended reading: *Head First C* Chapter 2.

Chapter 4 is about file systems and data streams. Recommended reading: *Head First C* Chapter 3.

Chapter 5 describes how numbers, letters, and other values are encoded, and presents the bitwise operators.

Chapter 6 explains how to use dynamic memory management, and how it works. Recommended reading: *Head First C* Chapter 6.

Chapter 7 is about caching and the memory hierarchy.

Chapter 8 is about multitasking and scheduling.

Chapter 9 is about POSIX threads and mutexes. Recommended reading: *Head First C* Chapter 12 and *Little Book of Semaphores* Chapters 1 and 2.

Chapter 10 is about POSIX condition variables and the producer/consumer problem. Recommended reading: *Little Book of Semaphores* Chapters 3 and 4.

Chapter 11 is about using POSIX semaphores and implementing semaphores in C.

A note on this draft

The current version of this book is an early draft. While I am working on the text, I have not yet included the figures. So there are a few places where, I'm sure, the explanation will be greatly improved when the figures are ready.

0.1 Using the code

Example code for this book is available from <https://github.com/AllenDowney/ThinkOS>. Git is a version control system that allows you to keep track of the files that make up a project. A collection of files under Git's control is called a **repository**. GitHub is a hosting service that provides storage for Git repositories and a convenient web interface.

The GitHub homepage for my repository provides several ways to work with the code:

- You can create a copy of my repository on GitHub by pressing the **Fork** button. If you don't already have a GitHub account, you'll need to create one. After forking, you'll have your own repository on GitHub